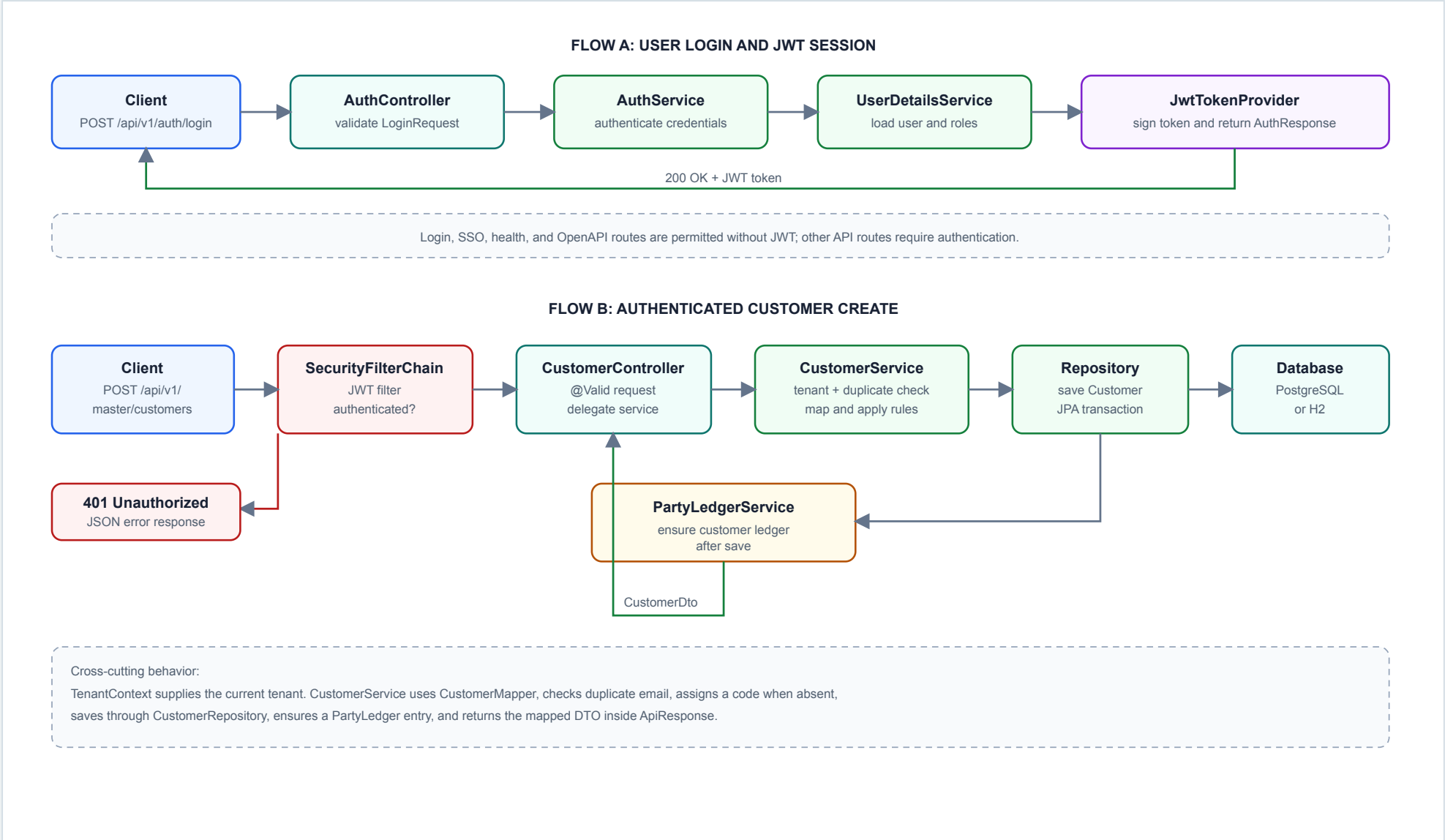


ERP Platform Request Flow

Code-grounded flow for authentication and an authenticated customer operation



Source evidence: SecurityConfig.java, JwtAuthFilter.java, AuthController.java, CustomerController.java, CustomerService.java, CustomerRepository.java, application*.properties.

Flow Notes

Decision points and source mapping

1. Authentication Entry

Public login is mapped by `AuthController.login()`. The request is validated, client IP is resolved, and `AuthService.login()` authenticates the user before `JwtTokenProvider` signs the session token.

2. Request Protection

`SecurityConfig` configures stateless sessions and inserts `JwtAuthFilter` before the `username/password` filter. Any non-public route must be authenticated.

3. API Boundary

`CustomerController` owns the HTTP contract, pagination parameters, validation, status codes, and response wrapping. It does not implement persistence or business rules.

4. Tenant-Aware Business Logic

`CustomerService` reads the current tenant, rejects duplicate active email addresses, maps DTOs to entities, assigns a generated customer code, and coordinates ledger creation.

5. Persistence

`CustomerRepository` extends Spring Data JPA. The service runs in a transaction, and Hibernate persists the entity into PostgreSQL in production or H2 in local/test profiles.

6. Failure Paths

- Missing or invalid token: JSON 401 response.
- Validation failure: Spring validation error handling.
- Duplicate email: conflict exception.
- Missing customer: not-found exception.

Representative Modules

The same controller-service-repository pattern is repeated across accounting, sales, purchase, inventory, agri, production, HR, payroll, reports, and other business packages.

Background Flow

Scheduled jobs bypass an HTTP client and invoke application services in-process. Field synchronization optionally calls Open-Meteo or configured satellite providers and persists field data.

Primary Source Locations

- **Bootstrap:** `src/main/java/com/erp/platform/ErpPlatformApplication.java`
- **Security:** `src/main/java/com/erp/platform/security/`
- **Authentication:** `src/main/java/com/erp/platform/modules/auth/`
- **Customer flow:** `src/main/java/com/erp/platform/modules/master/{controller,service,repository}/`

- **Database:** src/main/resources/db/migration/ and application*.properties